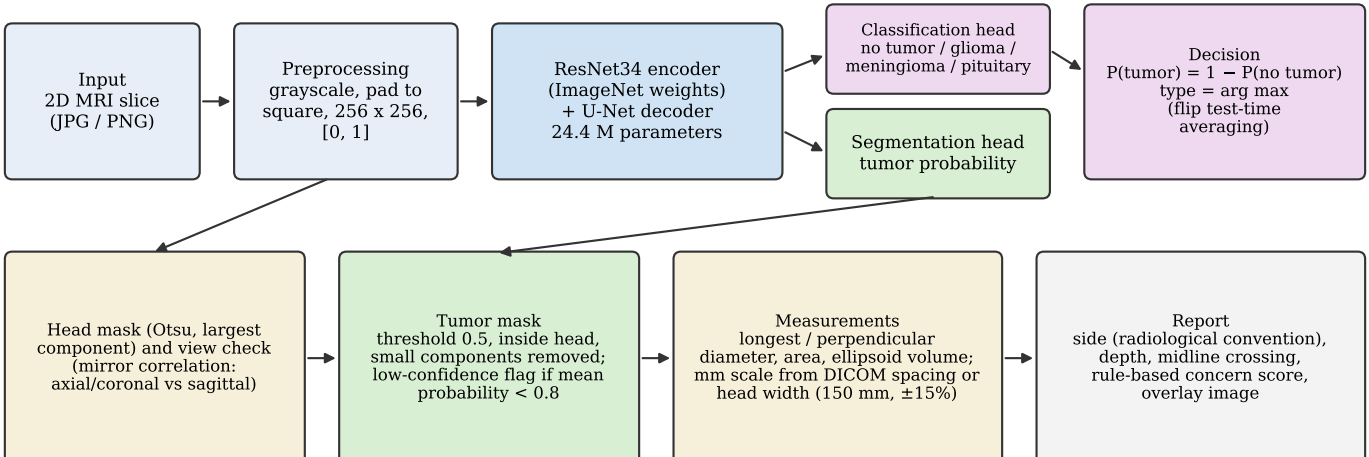
Input
2D MRI slice
(JPG / PNG)
Preprocessing
grayscale, pad to
square, 256 x 256,
[0, 1]
ResNet34 encoder
(ImageNet weights)
+ U-Net decoder
24.4 M parameters
Classification head
no tumor / glioma /
meningioma / pituitary
Segmentation head
tumor probability
Decision
P(tumor) = 1 − P(no tumor)
type = arg max
(flip test-time
averaging)
Head mask (Otsu, largest
component) and view check
(mirror correlation:
axial/coronal vs sagittal)
Tumor mask
threshold 0.5, inside head,
small components removed;
low-confidence flag if mean
probability < 0.8
Measurements
longest / perpendicular
diameter, area, ellipsoid volume;
mm scale from DICOM spacing or
head width (150 mm, ±15%)
Report
side (radiological convention),
depth, midline crossing,
rule-based concern score,
overlay image